

V5__Create_Enterprise_Schema.sql

Path: backend-service/src/main/resources/db/migration/V5__Create_Enterprise_Schema.sql

Purpose: Clean slate migration - drops old tables and creates enterprise-grade schema

SQL Migration File (Copy Entire Content)

```
-- =====
-- Priority 1: Enterprise-Grade Schema Migration
-- Strategy: Clean Slate (Drop old + Create new)
-- Tables: api_logs, anomaly_detections, system_metrics, distributed_traces, alert_rules
-- =====
```

```
-- =====
-- PART 1: Cleanup (Drop old tables if they exist)
-- =====
```

```
DROP TABLE IF EXISTS public.anomalies CASCADE;
DROP TABLE IF EXISTS public.metrics CASCADE;
DROP TABLE IF EXISTS public.traces CASCADE;
DROP TABLE IF EXISTS public.alerts CASCADE;
DROP TABLE IF EXISTS public.anomalyscores CASCADE;
```

```
-- =====
-- PART 2: Create api_logs Table (Primary data source)
-- Purpose: Store ALL API request/response data for ML analysis
-- =====
```

```
CREATE TABLE api_logs (
-- Primary Key
id BIGSERIAL PRIMARY KEY,
```

```
-- API Request Info
endpoint VARCHAR(500) NOT NULL,
http_method VARCHAR(10) NOT NULL CHECK (http_method IN ('GET', 'POST', 'P
status_code INTEGER NOT NULL CHECK (status_code >= 100 AND status_code <
response_time_ms BIGINT NOT NULL CHECK (response_time_ms >= 0),

-- Request/Response Size (bytes)
request_size_bytes BIGINT,
response_size_bytes BIGINT,
```

```

-- System Metrics
cpu_usage_percent DOUBLE PRECISION CHECK (cpu_usage_percent >= 0 AND cpu_usage_percent <= 100),
memory_usage_percent DOUBLE PRECISION CHECK (memory_usage_percent >= 0 AND memory_usage_percent <= 100),
disk_io_bytes BIGINT,
network_io_bytes BIGINT,

-- Error Tracking
error_rate DOUBLE PRECISION DEFAULT 0.0 CHECK (error_rate >= 0 AND error_rate <= 1),
error_count INTEGER DEFAULT 0,
error_message TEXT,
stack_trace TEXT,

-- Request Context
request_count INTEGER DEFAULT 1 CHECK (request_count > 0),
user_id VARCHAR(255),
ip_address INET,
user_agent TEXT,

-- Request/Response Bodies (flexible JSON)
request_body JSONB,
response_body JSONB,
request_headers JSONB,
response_headers JSONB,

-- Distributed Tracing
trace_id VARCHAR(255),
span_id VARCHAR(255),
parent_span_id VARCHAR(255),

-- Service Context
service_name VARCHAR(255) NOT NULL,
service_version VARCHAR(50),
environment VARCHAR(50) DEFAULT 'production' CHECK (environment IN ('development', 'production', 'staging')),

-- Temporal Features (for ML)
hour_of_day INTEGER CHECK (hour_of_day >= 0 AND hour_of_day < 24),
day_of_week INTEGER CHECK (day_of_week >= 1 AND day_of_week <= 7),
is_weekend BOOLEAN,

```

```

is_business_hours BOOLEAN,

-- Processing Status
is_processed BOOLEAN NOT NULL DEFAULT FALSE,
processed_at TIMESTAMP,
anomaly_detection_id BIGINT,
ml_service_version VARCHAR(50),

-- Audit Fields (Enterprise Standard)
created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
created_by VARCHAR(255),
updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
updated_by VARCHAR(255),
deleted_at TIMESTAMP,
deleted_by VARCHAR(255),

-- Metadata (for extensibility)
metadata JSONB

```

```
);
```

```
-- Table Comment
```

```

COMMENT ON TABLE api_logs IS 'Primary data source: Stores ALL API request/response
data for ML-based anomaly detection (Priority 1 Enterprise Integration)';
COMMENT ON COLUMN api_logs.trace_id IS 'Distributed trace identifier for request
correlation';
COMMENT ON COLUMN api_logs.is_processed IS 'Flag indicating if this log has been
analyzed by ML service';
COMMENT ON COLUMN api_logs.metadata IS 'Flexible JSON field for additional context
(custom fields, tags)';

```

```
-- Performance Indexes
```

```

CREATE INDEX idx_api_logs_endpoint ON api_logs(endpoint) WHERE deleted_at IS NULL;
CREATE INDEX idx_api_logs_status_code ON api_logs(status_code) WHERE deleted_at IS
NULL;
CREATE INDEX idx_api_logs_created_at ON api_logs(created_at DESC) WHERE deleted_at IS
NULL;
CREATE INDEX idx_api_logs_trace_id ON api_logs(trace_id) WHERE deleted_at IS NULL;

```

```
-- Composite Indexes (common query patterns)
```

```

CREATE INDEX idx_api_logs_endpoint_created ON api_logs(endpoint, created_at DESC)
WHERE deleted_at IS NULL;
CREATE INDEX idx_api_logs_service_created ON api_logs(service_name, created_at DESC)
WHERE deleted_at IS NULL;

```

```

-- Processing Index (batch operations)
CREATE INDEX idx_api_logs_unprocessed ON api_logs(is_processed, created_at)
WHERE deleted_at IS NULL AND is_processed = FALSE;

-- BRIN Index (very efficient for time-series data on large tables)
CREATE INDEX idx_api_logs_created_at_brin ON api_logs USING BRIN(created_at);

-- =====
-- PART 3: Create anomaly_detections Table (ML Predictions)
-- Purpose: Store ML model predictions and anomaly scores
-- =====

CREATE TABLE anomaly_detections (
-- Primary Key
id BIGSERIAL PRIMARY KEY,

-- Link to source log
api_log_id BIGINT REFERENCES api_logs(id) ON DELETE SET NULL,

-- API Info (denormalized for fast queries)
endpoint VARCHAR(500) NOT NULL,
http_method VARCHAR(10) NOT NULL,

-- ML Model Scores (All three algorithms)
msif_lstm_score DOUBLE PRECISION NOT NULL CHECK (msif_lstm_score >= 0 AND msif_lstm_score <= 1),
ple_gru_score DOUBLE PRECISION NOT NULL CHECK (ple_gru_score >= 0 AND ple_gru_score <= 1),
hybrid_ensemble_score DOUBLE PRECISION NOT NULL CHECK (hybrid_ensemble_score >= 0 AND hybrid_ensemble_score <= 1),

-- Confidence & Severity
confidence_score DOUBLE PRECISION NOT NULL CHECK (confidence_score >= 0 AND confidence_score <= 1),
severity_level VARCHAR(50) NOT NULL CHECK (severity_level IN ('LOW', 'MEDIUM', 'HIGH')),
anomaly_type VARCHAR(100), -- response_time, error_rate, resource_usage, network_latency

-- ML Model Info
fusion_method VARCHAR(100) NOT NULL, -- weighted_ensemble, voting, stack
ml_model_version VARCHAR(50),
ml_processing_time_ms BIGINT,

-- Status Management
status VARCHAR(50) NOT NULL DEFAULT 'ACTIVE' CHECK (status IN ('ACTIVE', 'PENDING', 'REMOVED')),
is_acknowledged BOOLEAN DEFAULT FALSE,
acknowledged_by VARCHAR(255),

```

```

acknowledged_at TIMESTAMP,
acknowledgement_note TEXT,

-- Resolution
is_resolved BOOLEAN DEFAULT FALSE,
resolved_by VARCHAR(255),
resolved_at TIMESTAMP,
resolution_note TEXT,

-- False Positive Handling
is_false_positive BOOLEAN DEFAULT FALSE,
marked_false_positive_by VARCHAR(255),
marked_false_positive_at TIMESTAMP,

-- Distributed Tracing
trace_id VARCHAR(255),

-- Service Context
service_name VARCHAR(255),
environment VARCHAR(50),

-- Audit Fields (Enterprise Standard)
created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
created_by VARCHAR(255),
updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
updated_by VARCHAR(255),
deleted_at TIMESTAMP,
deleted_by VARCHAR(255),

-- Metadata
additional_context JSONB

```

```
);
```

```

-- Table Comment
COMMENT ON TABLE anomaly_detections IS 'ML anomaly detection results: Stores
predictions from MSIF-LSTM, PLE-GRU, and Hybrid-Ensemble models (Priority 1 Enterprise
Integration)';
COMMENT ON COLUMN anomaly_detections.hybrid_ensemble_score IS 'Final score from
weighted ensemble of all three models (primary scoring metric)';

```

```
COMMENT ON COLUMN anomaly_detections.status IS 'Lifecycle status: ACTIVE (ongoing),  
RESOLVED (handled), FALSE_POSITIVE (incorrect detection)';
```

```
-- Performance Indexes
```

```
CREATE INDEX idx_anomaly_detections_endpoint ON anomaly_detections(endpoint)  
WHERE deleted_at IS NULL;  
CREATE INDEX idx_anomaly_detections_severity ON anomaly_detections(severity_level)  
WHERE deleted_at IS NULL;  
CREATE INDEX idx_anomaly_detections_status ON anomaly_detections(status) WHERE  
deleted_at IS NULL;  
CREATE INDEX idx_anomaly_detections_created_at ON anomaly_detections(created_at  
DESC) WHERE deleted_at IS NULL;  
CREATE INDEX idx_anomaly_detections_trace_id ON anomaly_detections(trace_id)  
WHERE deleted_at IS NULL;  
CREATE INDEX idx_anomaly_detections_api_log_id ON anomaly_detections(api_log_id)  
WHERE deleted_at IS NULL;
```

```
-- Composite Indexes (dashboard queries)
```

```
CREATE INDEX idx_anomaly_detections_severity_status ON  
anomaly_detections(severity_level, status, created_at DESC)  
WHERE deleted_at IS NULL;
```

```
-- Hot Query Index (unacknowledged critical anomalies)
```

```
CREATE INDEX idx_anomaly_detections_critical_unack ON anomaly_detections(created_at  
DESC)  
WHERE deleted_at IS NULL AND is_acknowledged = FALSE AND severity_level IN  
( 'CRITICAL', 'HIGH');
```

```
-- =====
```

```
-- PART 4: Create system_metrics Table (Time-series data)
```

```
-- Purpose: Store system/service metrics for correlation with anomalies
```

```
-- =====
```

```
CREATE TABLE system_metrics (  
id BIGSERIAL PRIMARY KEY,
```

```
-- Link to API log (if available)
```

```
api_log_id BIGINT REFERENCES api_logs(id) ON DELETE SET NULL,
```

```
-- Service Info
```

```
service_name VARCHAR(255) NOT NULL,
```

```
endpoint VARCHAR(500),
```

```
-- Metrics (from original metrics table)
```

```
cpu_usage_percent DOUBLE PRECISION,
```

```
memory_usage_percent DOUBLE PRECISION,
```

```
disk_io_bytes BIGINT,
```

```

network_io_bytes BIGINT,
response_time_ms BIGINT,
request_count INTEGER,
error_rate DOUBLE PRECISION,

-- Timestamp
metric_timestamp TIMESTAMP NOT NULL,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

```

```
);
```

```
-- Table Comment
```

```
COMMENT ON TABLE system_metrics IS 'Time-series system metrics: Stores CPU, memory,
disk, network metrics for analysis and correlation';
```

```
-- Performance Indexes
```

```
CREATE INDEX idx_system_metrics_timestamp ON system_metrics(metric_timestamp
DESC);
```

```
CREATE INDEX idx_system_metrics_service ON system_metrics(service_name,
metric_timestamp DESC);
```

```
CREATE INDEX idx_system_metrics_api_log_id ON system_metrics(api_log_id) WHERE
api_log_id IS NOT NULL;
```

```

-- =====
-- PART 5: Create distributed_traces Table (Distributed Tracing)
-- Purpose: Store distributed tracing data for request correlation
-- =====

```

```
CREATE TABLE distributed_traces (
id BIGSERIAL PRIMARY KEY,
```

```
-- Trace Info
```

```
trace_id VARCHAR(255) NOT NULL,
```

```
span_id VARCHAR(255) NOT NULL,
```

```
parent_span_id VARCHAR(255),
```

```
-- Service Info
```

```
service_name VARCHAR(255) NOT NULL,
```

```
operation_name VARCHAR(255),
```

```
-- Timing
```

```
start_time TIMESTAMP NOT NULL,
```

```
duration_ms BIGINT NOT NULL,
```

```

-- Status
status_code INTEGER,
is_error BOOLEAN DEFAULT FALSE,
error_message TEXT,

-- Tags (flexible metadata)
tags JSONB,

-- Audit
created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

);

-- Table Comment
COMMENT ON TABLE distributed_traces IS 'Distributed tracing data: Stores spans for
request tracing across services';

-- Performance Indexes
CREATE INDEX idx_distributed_traces_trace_id ON distributed_traces(trace_id);
CREATE INDEX idx_distributed_traces_service ON distributed_traces(service_name,
start_time DESC);
CREATE INDEX idx_distributed_traces_parent ON distributed_traces(parent_span_id)
WHERE parent_span_id IS NOT NULL;
CREATE INDEX idx_distributed_traces_created_at ON distributed_traces(created_at DESC);

-- =====
-- PART 6: Create alert_rules Table (Alert Configuration)
-- Purpose: Store alert rules and notification configuration
-- =====

CREATE TABLE alert_rules (
id BIGSERIAL PRIMARY KEY,

-- Alert Config
alert_name VARCHAR(255) NOT NULL,
alert_description TEXT,

-- Condition
condition_type VARCHAR(100) NOT NULL CHECK (condition_type IN ('threshold', 'condition')),
condition_expression TEXT NOT NULL,
threshold_value DOUBLE PRECISION,

```



```

-- Severity
severity_level VARCHAR(50) NOT NULL CHECK (severity_level IN ('LOW', 'MEDI

-- Status
is_enabled BOOLEAN NOT NULL DEFAULT TRUE,

-- Notification
notification_channels JSONB, -- ["email", "slack", "pagerduty", "webhook"]
notification_recipients JSONB,

-- Audit
created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
created_by VARCHAR(255),
updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
updated_by VARCHAR(255),
deleted_at TIMESTAMP,
deleted_by VARCHAR(255)

);

-- Table Comment
COMMENT ON TABLE alert_rules IS 'Alert rules: Stores alert configuration, thresholds, and
notification channels';

-- Performance Indexes
CREATE INDEX idx_alert_rules_enabled ON alert_rules(is_enabled) WHERE deleted_at IS
NULL AND is_enabled = TRUE;
CREATE INDEX idx_alert_rules_severity ON alert_rules(severity_level) WHERE deleted_at IS
NULL;
CREATE INDEX idx_alert_rules_created_at ON alert_rules(created_at DESC) WHERE
deleted_at IS NULL;

-- =====
-- PART 7: Create Views (Optional - for common queries)
-- =====

-- Recent anomalies view
CREATE OR REPLACE VIEW v_recent_anomalies AS
SELECT
a.id,
a.endpoint,
a.http_method,
a.severity_level,
a.hybrid_ensemble_score,
a.status,

```

```
a.is_acknowledged,  
a.created_at  
FROM anomaly_detections a  
WHERE a.deleted_at IS NULL  
ORDER BY a.created_at DESC  
LIMIT 100;
```

```
COMMENT ON VIEW v_recent_anomalies IS 'Recent 100 anomalies for dashboard';
```

```
-- Anomaly statistics view
```

```
CREATE OR REPLACE VIEW v_anomaly_stats AS  
SELECT  
DATE(a.created_at) as date,  
a.severity_level,  
COUNT(*) as count,  
AVG(a.hybrid_ensemble_score) as avg_score,  
MAX(a.hybrid_ensemble_score) as max_score,  
MIN(a.hybrid_ensemble_score) as min_score,  
SUM(CASE WHEN a.is_acknowledged THEN 1 ELSE 0 END) as acknowledged_count  
FROM anomaly_detections a  
WHERE a.deleted_at IS NULL  
GROUP BY DATE(a.created_at), a.severity_level  
ORDER BY date DESC;
```

```
COMMENT ON VIEW v_anomaly_stats IS 'Daily anomaly statistics by severity';
```

```
-- =====  
-- PART 8: Grant Permissions  
-- =====
```

```
GRANT SELECT, INSERT, UPDATE, DELETE ON api_logs TO api_monitor;  
GRANT SELECT, INSERT, UPDATE, DELETE ON anomaly_detections TO api_monitor;  
GRANT SELECT, INSERT, UPDATE, DELETE ON system_metrics TO api_monitor;  
GRANT SELECT, INSERT, UPDATE, DELETE ON distributed_traces TO api_monitor;  
GRANT SELECT, INSERT, UPDATE, DELETE ON alert_rules TO api_monitor;  
GRANT SELECT ON v_recent_anomalies TO api_monitor;  
GRANT SELECT ON v_anomaly_stats TO api_monitor;
```

```
-- Grant sequence permissions for auto-increment
```

```
GRANT USAGE, SELECT ON ALL SEQUENCES IN SCHEMA public TO api_monitor;
```

```

--
=====
=====
-- PART 9: Migration Summary
--
=====
=====
-- This migration creates an enterprise-grade schema with:
-- ✓ Proper naming conventions (snake_case, descriptive
names)
-- ✓ Audit fields (created_at, created_by, deleted_at for
soft deletes)
-- ✓ Performance indexes (single, composite, BRIN)
-- ✓ Constraints (CHECK, NOT NULL, FOREIGN KEY)
-- ✓ Table comments and column documentation
-- ✓ Support for multi-tenancy (can add tenant_id later)
-- ✓ Views for common queries
-- ✓ Proper permissions for api_monitor user

-- Old tables dropped: anomalies, metrics, traces, alerts,
anomalyscores

-- Data migration: Old data is NOT preserved (clean slate approach)
-- =====

```

▮ How to Apply This Migration

Step 1: Create the file

```
cd ~/AI-Powered-API-Monitoring-And-Multi-Source-Anomaly-Identification-Model-For-
Distributed-Platforms/backend-service
```

```
cat > src/main/resources/db/migration/V5__Create_Enterprise_Schema.sql <<
'EOFMIGRATION'
-- [Paste the entire SQL content from above]
EOFMIGRATION
```

Step 2: Verify file was created

```
ls -lh src/main/resources/db/migration/V*.sql
```

Expected output:

```
V1__Create_anomaly_tables.sql
V2__Create_alerts_table.sql
V3__Create_metrics_anomalies_tables.sql
V4__Create_traces_table.sql
V5__Create_Enterprise_Schema.sql ← NEW FILE
```

Step 3: Build and apply migration

Build (Flyway will run migrations automatically)

```
./gradlew clean build -x test
```

Verify migration succeeded

```
PGPASSWORD='api_monitor_pwd' psql -h localhost -U api_monitor -d api_monitoring -c "\dt"
```

Expected tables:

Schema	Name	Type	Owner
public	alert_rules	table	api_monitor ✓ NEW
public	anomaly_detections	table	api_monitor ✓ NEW
public	api_logs	table	api_monitor ✓ NEW
public	distributed_traces	table	api_monitor ✓ NEW
public	system_metrics	table	api_monitor ✓ NEW
public	flyway_schema_history	table	postgres

(6 rows)

✓ Schema Structure Summary

Table	Purpose	Rows	Key Fields
api_logs	Raw API logs	0 (fresh)	endpoint, method, status_code, response_time_ms
anomaly_detections	ML predictions	0 (fresh)	msif_lstm_score, ple_gru_score, hybrid_ensemble_score
system_metrics	Time-series metrics	0 (fresh)	cpu_usage_percent, memory_usage_percent
distributed_traces	Tracing spans	0 (fresh)	trace_id, span_id, duration_ms
alert_rules	Alert config	0 (fresh)	alert_name, severity_level, condition_type

▮ Next: Update Java Entities

After this migration is applied, you'll need to update:

1. **LogRecord.java** → Map to api_logs table
2. **AnomalyRecord.java** → Map to anomaly_detections table
3. Update repositories to use new table names

Ready for the next file? ▮